

ANSI STANDARDS

POWER IN DIOPTERS		TOLERANCE
SPHERE POWER:	0.00 TO 6.50 ABOVE 6.50	+ OR - 0.13 D + OR - 2.%
CYLINDER POWER:	0.00 TO 2.00 2.12 TO 4.50 ABOVE 4.50	+ OR - 0.13 D + OR - 0.15 D + OR - 4.%
CYLINDER AXIS:	0.125 TO 0.375 0.50 TO 0.75 0.875 TO 1.50 ABOVE 1.50	+ OR - 7 degrees + OR - 5 degrees + OR - 3 degrees + OR - 2 degrees
MULTIFOCAL ADD: (POWER IN DISTANCE)	0.00 TO 8.00 ABOVE 8.00	+ OR - 0.13 D + OR - 0.18 D
SEGMENT HEIGHT:		+ OR - 1 mm
BASE CURVE:		+ OR - 0.75 D
<u>PRISMATIC IMBALANCE:</u>		
(TOTAL MOUNTED PAIR): HORIZONTAL= 0.667 OR MRPs WITHIN 2.5mm		
(EACH LENS): HORIZONTAL= 0.333 VERTICAL= 0.333 OR MRPs WITHIN 1.mm		

INDEX OF REFRACTION:

<u>MATERIAL</u>	<u>INDEX</u>
AIR-----	1.000
VITREOUS HUMOR-----	1.333
CORNEA-----	1.370
CRYSTALLINE LENS-----	1.420
CR-39-----	1.490
CROWN GLASS-----	1.523
TRIVEX-----	1.530
POLYCARBONATE-----	1.589
BARIUM GLASS-----	1.602
TITANIUM GLASS-----	1.701

BASE CURVE CHART

<u>POWER RANGE</u>	<u>BASE CURVE</u>
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+5.50 TO +8.75	+10.00
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+2.25 TO +5.25	+8.00
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PLANO TO +2.00	+6.00
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PLANO TO -1.75	+6.00
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-2.00 TO -4.50	+4.00
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-4.75 TO -8.00	+2.00
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-8.00 TO -9.00	+1.00
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-9.00 AND ABOVE	PLANO
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NOTE: Most popular base curve used is a + 6.00

LENS THICKNESS

Lens thickness is determined by several factors:

- The customer's prescription.
- The lens material selected.
- The frame size.
- The decentration.

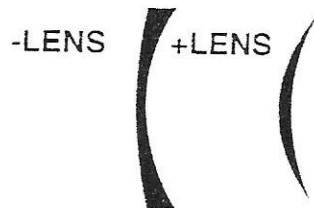
Of these four factors, it's obvious that we can have little influence on prescription, and while some lens materials can be made thinner than others, the customer or optician expecting profound thickness improvement by switching lens materials is bound for disappointment. The two thickness factors over which we have the most control are frame size and decentration.

Lens thickness is first determined by power. A minus lens is by definition, thin in the center and thicker at the edge. Conversely, the center thickness of a plus lens is greater than its edge thickness.

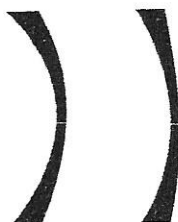
Optimum center thickness for minus lenses ranges from 1.5 mm for polycarbonate to 2.0 for regular plastic lenses.

Optimum edge thickness for plus ranges from 1.5 mm to 2.5 mm depending on frame type ("rimless" style frames require a thicker edge to accommodate the groove).

Once thickness at the thinnest point is established, the edge thickness (if the lens is a minus), or the center thickness (if a plus), will then be influenced by lens prescription, frame size, and decentration.



Though the minimum thickness remains constant, thickness will increase as the power of the prescription increases. Therefore, of two lenses having the same minimum thickness, the lens with the stronger prescription will be thicker.



Frame size and decentration influence lens thickness by determining lens blank diameter. The larger the lens diameter, the greater the thickness. In lenses of equal power, the larger lens will always be thicker.

SEG HEIGHTS

The segment height is the distance (in millimeters) from the lowest point of the eyewire to the top of the segment (see figure 1), not from the bottom of the segment (see figure 2).

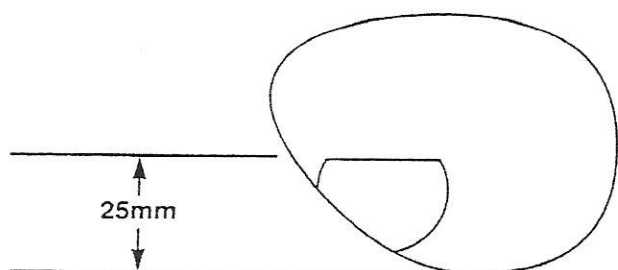


FIGURE 1 - CORRECT

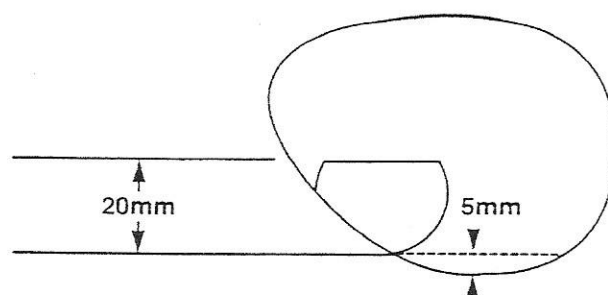


FIGURE 2 - NOT CORRECT

HOW TO MEASURE THE CUSTOMER FOR SEGMENT HEIGHT

Most typical and successful segment positions for:

Progressives: - To the center of the pupil (A)

Trifocals: - To the bottom of the pupil (B)

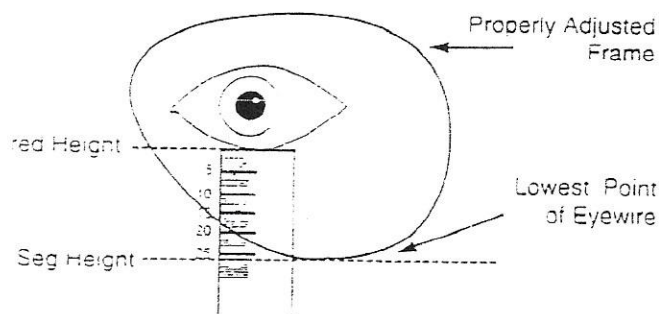
All other Bifocals - To the lower lid (C)

PROCEDURE:

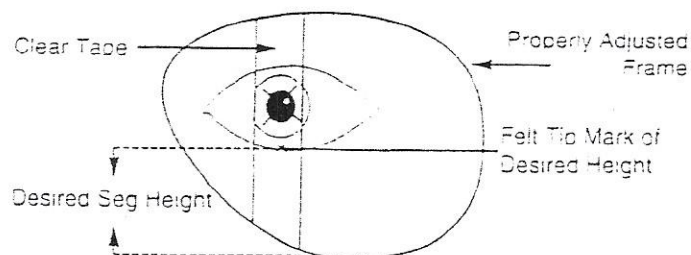
- Use a PD ruler and a properly adjusted frame of choice.
- Position yourself, eye to eye with customer at a distance of approximately 16 inches.
- With frame of choice properly adjusted, place the zero reading of the PD ruler at the desired location of the segment top.
- Look down to lowest point of eye wire. Reading on PD ruler at that point is the seg height. Note: If at time of measurement frame is not resting where it will normally be worn, the reading will be inaccurate.

NOTE: An alternative to the above would be to place a piece of clear tape vertically across properly adjusted frame. With frame on, use felt tip pen to mark desired height. Remove frame to measure from lowest point on eye wire to reference mark. This technique is particularly helpful for locating pupillary heights when fitting progressives since it automatically isolates both horizontal and vertical co-ordinates.

USE OF P.D. RULE



WITH CLEAR TAPE





A

planoconcave



B

planoconvex



C

biconcave



D

biconvex



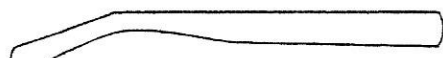
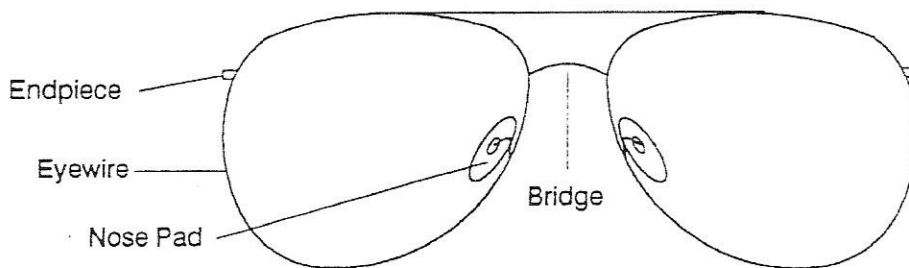
E

meniscus

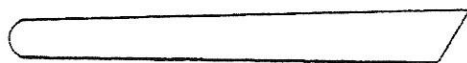


F

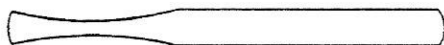
meniscus



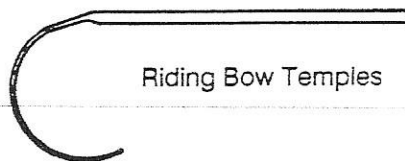
Skull Temples



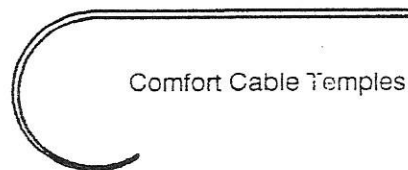
Library Temples



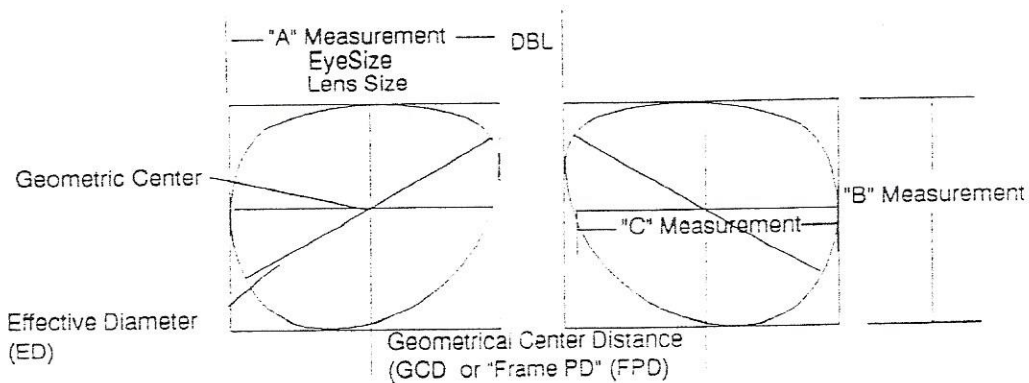
Convertible Temples



Riding Bow Temples

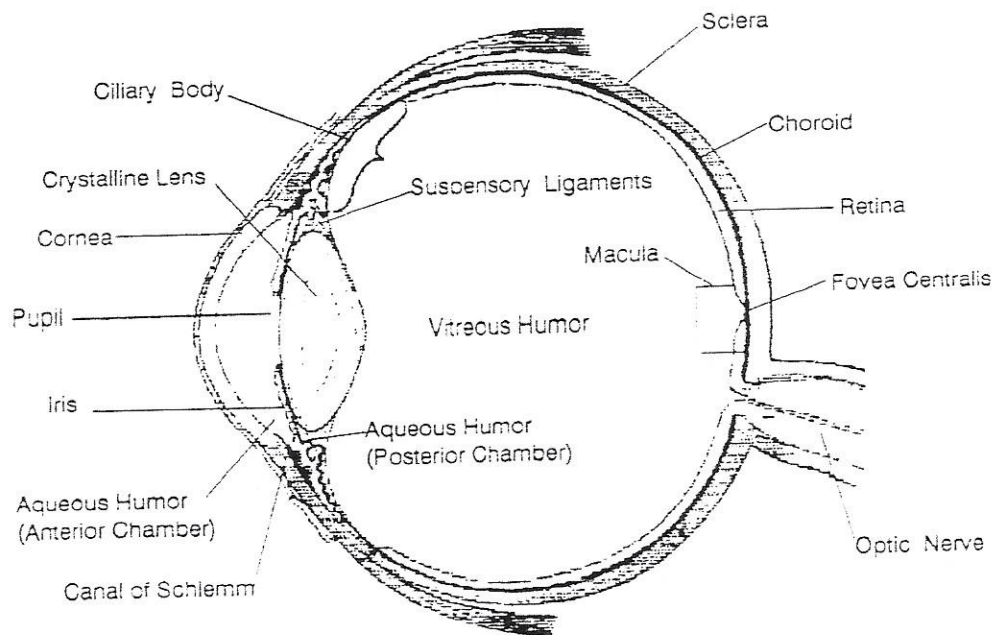
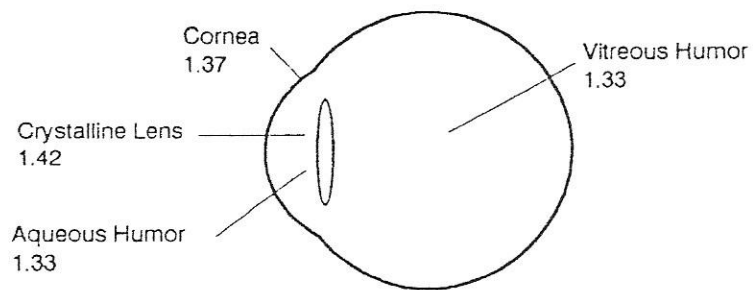


Comfort Cable Temples



Box System of Measurement

CONES	RODS
Day Vision	Night Vision
Detail	Peripheral Vision
Color	Black and White



DISPENSING TIPS

If the frame sits on your patient with the right side high, simply push the right temple up.

If the frame sits high on the left side, simply push the left temple up.

If the frame front sits too close to the patient's left side of the face, simply bring in the left temple.

If the frame front sits too close to the patient's right side of the face, simply bring in the right temple.

To increase the pantoscopic tilt, simply bend both temples down.

To decrease the pantoscopic tilt, simply bend both temples up.

Increase pantoscopic is the same as decrease retroscopic.

Decrease pantoscopic is the same as increase retroscopic.

An oval shaped face can wear almost any shape frame.

Round faces should avoid round frames.

For high prescriptions use smaller rounder shapes.

QUICK REFERENCE GUIDE ON MATH FORMULA'S

FRAME P.D. - PATIENTS P.D. DIVIDED BY 2 = DECENTRATION.

PRISM x 10 DIVIDED BY THE LENS DIOPTER = HOW TO CREATE PRISM.

d x D DIVIDED BY 10 = HOW MUCH UNWANTED PRISM IS FOUND.

VERTEX MATH:

_____ STEP 1. 1000 DIVIDED BY D

_____ STEP 2. ADD OR SUBTRACT AMOUNT OF MOVEMENT.

_____ STEP 3. 1000 DIVIDED BY THAT ANSWER.

TABLE OF ABBREVEATIONS

DIOPTERS	=====	D
DECENTRATION	=====	d
PRISM	=====	Δ
PLUS LENS MOVED IN, SUBTRACT	=====	P.I.S.
MINUS LENS MOVED IN, ADD	=====	M.I.A.
RIGHT EYE	=====	O.D.
LEFT EYE	=====	O.S.
BOTH EYES	=====	O.U.
MILLIMETERS	=====	MM
CENTIMETERS	=====	CM
METERS	=====	M

TO CALCULATE THE FOCAL LENGTH, IS A SIMPLE ONE STEP DIVISION.

+5.00 WHAT IS THE FOCAL LENGTH IN M _____
CM _____
MM _____

M=1

CM=100

MM=1000

FOR M DIVIDE 1 BY 5 = .20

FOR CM DIVIDE 100 BY 5 = 20

FOR MM DIVIDE 1000 BY 5 = 200

Things to remember about prism.

- #1 When splitting prism, it is always necessary to keep the same direction of the base in the lens it was prescribed.

Example: original Rx. +1.00 - 1.00 x 134 4^Δ B.I.
+1.50 - 2.00 x 180

Rx. with prism properly split or distributed.

+1.00 - 1.00 x 134 2^Δ B.I.
+1.50 - 2.00 x 180 2^Δ B.O.

- #2 Remember to split the amount of prism equally between the two lenses.

Example: original Rx. +1.00 - 1.00 x 134 3^Δ B.O.
+1.50 - 2.00 x 180

Rx. with prism equally distributed.

+1.00 - 1.00 x 134 1.5^Δ B.O.
+1.50 - 2.00 x 180 1.5^Δ B.I.

- #3 Remember to split the prism so the two lenses are opposite direction of each other.

Example: if the Rx. reads B.O. in right lens then left lens will be B.I.

- #4 Remember that an Rx, can have more than one prism.

Example: original Rx. +1.00 - 1.00 x 134 3^Δ B.O.
+1.75 - 2.00 x 180 1^Δ B.U.

Correctly split example: +1.00 - 1.00 x 134 1.5^Δ B.O. .5^Δ B.D.
+1.75 - 2.00 x 180 1.5^Δ B.I. .5^Δ B.U.

TOPIC: SPLITTING PRISM (REDISTRIBUTING PRISM)

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It is possible to redistribute the prism between the two lenses without altering its effective power. For example, the following prescription:

O.D. +1.00 SPH 2^Δ B.U.
O.S. +1.00 SPH

Could be written:

O.D. +1.00 SPH 1^Δ B.U.
O.S. +1.00 SPH 1^Δ B.D.

Could be written:

O.D. +1.00 SPH
O.S. +1.00 SPH 2^Δ B.D.

Questions relating to splitting prism would always be asked of the prescribing doctor. An Optician would never do this without asking the doctor first. In each instance the same optical result is achieved. However the second example would distribute the weight and thickness more evenly than the first or third example. The result is a better fitting and more cosmetically appealing pair of eyeglasses. Should the Optician be given a prescription similar to these examples, it would be well within their scope to recommend splitting prism.

EXAMPLES:

ORIGINAL RX	PRISM REDISTRIBUTED
O.D. -2.00 -0.75 X030 4 ^Δ B.U. O.S. -1.75 -0.75 X010	O.D. -2.00 -0.75 X030 2 ^Δ B.U. O.S. -1.75 -0.75 X010 2 ^Δ B.D.
O.D. +0.50 -0.25 X180 O.S. +1.00 -0.50 X175 3 ^Δ B.U.	O.D. +0.50 -0.25 X180 1.5 ^Δ B.D. O.S. +1.00 -0.50 X175 1.5 ^Δ B.U.

What is a Prism. ^Δ stands for PRISM DIOPTER.

1. A plus lens has it's bases together resembling this



2. A minus lens has it's apex together, resembling this:



Plus lenses converge, while minus lenses diverge.

**TO CALCULATE THE MINIMUM BLANK SIZE FOR ANY
FRAME!**

$$(MBS) = (GCD - PD) + ED$$

GCD = GEOMETRICAL CENTER DISTANCE -OR- FRAME PD

PD = PATIENTS PD

ED = EFFECTIVE DIAMETER

DETERMINE THE THEORETICAL MINIMUM BLANK SIZE
FOR THE FOLLOWING.

#1 JOHN HAS A PD OF 63
HIS FRAME MEASURES A=50 B=42 ED=54 DBL=17
TEMPLE IS 145MM.

#2 SUE HAS A PD OF 57
HER FRAME MEASURES GCD=65 TEMPLE IS 135
DBL=18 ED=58

#3 JACK HAS A PD OF 67
HIS FRAME MEASURES A=56 DBL=16 ED=59

#4 JOE HAS A PD OF 64
HIS FRAME MEASURES A=50 DBL=18 ED=54 B=46

TOPIC: TRANSPOSITION

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TRANSPOSITION: Is to change a prescription that is written in PLUS (+) cylinder form into a MINUS (-) cylinder form.

WHY CHANGE FROM + to - cylinder form:

1. The majority of lenses are MINUS (-).
2. Some of the equipment used is in PLUS (+) cylinder form. EX:
+1.00 +0.50 X110

STEPS TO TRANSPOSITION:

1. The prescription has SPHERICAL, CYLINDER, AND AXIS. You will need to combine the SPH and CYL.
 - a. The rules of algebra apply here, NOTE if signs are the SAME you will always ADD, regardless if they are both PLUS (+) or both MINUS (-). EX:

+1.00	-1.00
<u>+1.25</u>	<u>-1.25</u>
+2.25	-2.25

Keep the signs of the numbers and add them together,

- b. If the signs are DIFFERENT, you will SUBTRACT and keep the sign of the larger number, EX:

+1.00
<u>-0.50</u>
+0.50

2. Change the SIGN of the CYLINDER from a PLUS to a MINUS, or visa versa. Remember the number stays the same, only the sign changes.

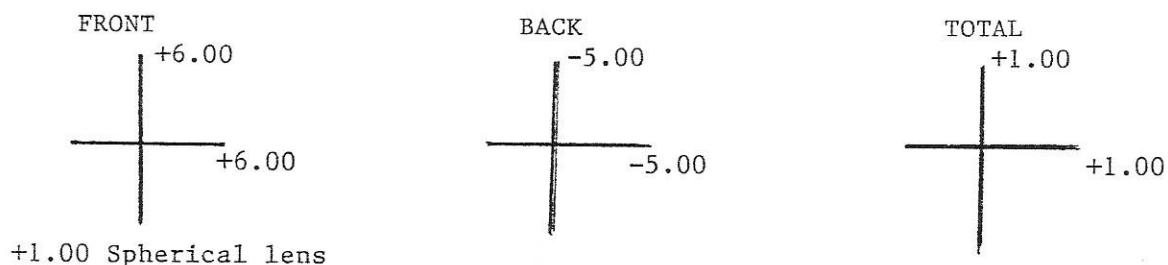
3. Rotate or move the AXIS 90 degrees, you can never have more than 180 degrees, so if you have 90 degrees or less ADD a full 90 degrees, if you have 91 degrees or more you must SUBTRACT a full 90 degrees. EX:

+1.00 +0.50 X90 becomes +1.50 -0.50 X180
-4.00 +1.00 X50 becomes -3.00 -1.00 X140

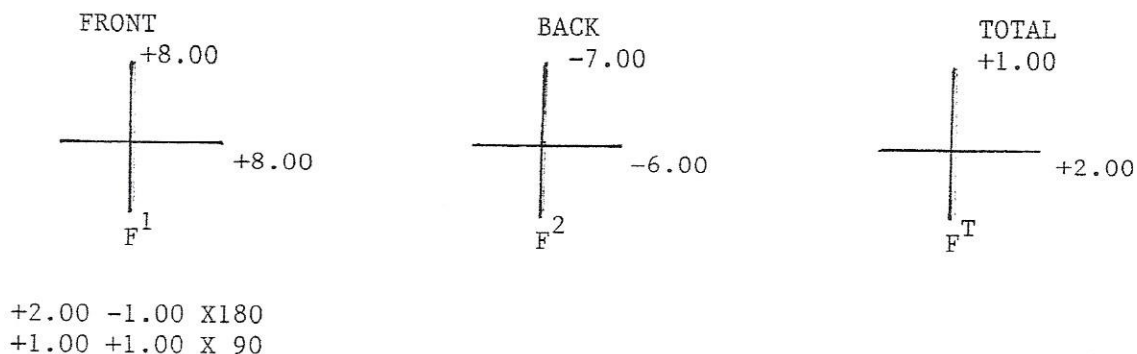
TOPIC: THE POWER CROSS (HANDOUT)

A POWER CROSS is a schematic representation of the two major meridians of a lens or lens surface. A spherical lens consists of two spherical surfaces, a front and a back surface to determine the total power of a lens simply combine the front and back surfaces algebraically by using the formula

$$F^1 + F^2 = F^T$$



When the lens obtains cylinder power from a difference in power between two front surface meridians (a toric front surface lens). The lens is said to be ground in PLUS CYLINDER FORM. If on the other hand, a lens has a cylinder component, but the cylinder power is a result of a difference in power between two back surface meridians (a toric back surface lens) it is a MINUS CYLINDER FORM lens. Power can be interpreted from information given on a power cross as follows:



TOPIC: POWERS IN VARIOUS MERIDIANS

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RULES:

To calculate the power of a lens in any meridian.

1. Be sure the RX (prescription) has its axis no more than 90 degrees from the unknown meridian. TRANSPOSE if needed

TRANSPOSE:

1. Combine (add) the SPH and CYL powers remembering if the signs are the same add them together. If the signs are different subtract keeping the sign of the higher number.
2. Change the cylinder sign from a plus (+) to a minus (-) or visa versa, do not change the number amount.
3. Rotate the axis a full 90 degrees. This means if the axis is 90 degrees or less you will add a full 90 degrees. If the axis is already 91 degrees or more, you will subtract a full 90 degrees.

EX: -1.00 -1.00 X90
-2.00 +1.00 X180

EX: +3.00 -2.00 X93
+1.00 +2.00 X003

2. Determine the number of degrees the unknown meridian is from the axis.
3. Compute the PERCENTAGE of the cylinder power in the unknown meridian.
4. Combine (add) the PERCENTAGE of the cylinder power to the sphere power. This equals the total power in the unknown meridian.

COMMON CYLINDER PERCENTAGES

<u>AWAY FROM AXIS</u>	<u>PERCENTAGE % OF CYLINDER POWER</u>
30 degrees	25% of cylinder power
45 degrees	50% of cylinder power
60 degrees	75% of cylinder power
90 degrees	100% of cylinder power

COMMIT THESE PERCENTAGES TO MEMORY

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Practice problems of powers in various meridians.

#1

+4.00 +4.00	-4.50 -3.50	x35 x125 ---- ----
Rx. _____ x35	at 95 _____	
Rx. _____ x125	at 05 _____	

#2

+8.00 +8.00	-9.50 -9.50	x90 x180 ---- ----
Rx. _____ x90	at 45 _____	
Rx. _____ x180	at 135 _____	

#3

+2.00 +2.00	-7.00 -7.00	x45 x135 ---- ----
Rx. _____ x45	at 90 _____	
Rx. _____ x135	at 180 _____	

#4

+6.00 +6.00	---- ----	x70 x160 -9.25 -9.25
Rx. _____ x70	at 100 _____	
Rx. _____ x123	at 179 _____	

Practice problems for optical math.

Transpose the following and name the refractive anomalies.

1. $+4.00 -4.00 \times 90$
2. $+0.75 +2.00 \times 87$
3. $+2.50 +1.25 \times 145$

The following glasses were made incorrectly, what would be the resulting prism and what direction would be the base?

	Patient p.d.	actual p.d.
4. $+1.00$ sph	58	63
5. -4.25 sph	61	64
6. -2.75 sph	57	54

If we want to create prism in the following lenses how much must each lens be moved and in what direction?

7. $+4.50$ with 2.00 diopters of base out prism?
8. -3.75 with 3.00 diopters of base out prism?

Determine the focal length of the following lenses in meters, cm and mm.

9. $+4.50$ sph

Remember that 1 meter equals 100 cm and 1000mm and 40 inches

10. -2.75 sph

11. If a lens has a virtual focal length of 20 cm what is the power?

Convert the following bifocal Rx into reading only.

12. $+1.00 -1.00 \times 90$
 $-1.00 -1.00 \times 89$
add 2.00
13. -1.00 sph
 $-1.25 -1.00 \times 90$
add o.d. 2.00
o.s. 2.50

14. Determine the minimum blank size for the following.
Frame info a=52 b= 40 ed=56 dbf=21 patients p.d.62 _____

Powers of various meridians.

15. +3.00 -1.00 x45 power at the 90 _____ and 180 _____
16. -2.75 +4.00 x60 power at the 45 _____ and 135 _____

Determine the theoretical minimum blank size for the following.

17. A=50 B= 45 DBL= 18 ED= 52 Temple= 135

Patients P.D. 64/60

Rx. +1.00 sph

+ 1.75 -1.00 x 90

For reading only.

- A. 56 mm
- B. 59 mm
- C. 61 mm
- D. 65 mm

18. How much would the seg drop be on this pair of glasses?

A=50 B=40 DBL=20 Temple 135 ED=54

Seg 18 mm

- A. 1.mm down
- B. 1.mm up
- C. 2.mm down
- D. 2.mm up
- E. none of the above

19. What would be the effective power you are looking through while looking through the intermediate segment of this trifocal lens?

Rx. o.d.+1.00 -1.00 x 180

o.s.+ 1.00 -1.00 x 180

ADD 3.00 o.u.

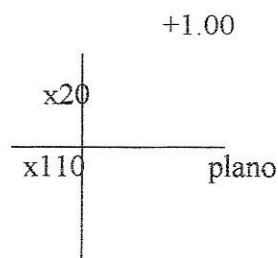
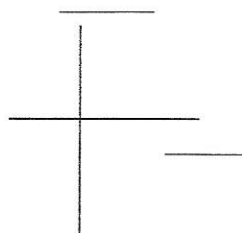
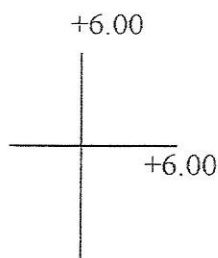
- A. +3.00
- B. +2.50
- C. +2.00
- D. +1.50
- E. None of the above

Powers in various meridians.

20. $-3.00 -1.00 \times 30$ power at the 90 _____ and 180 _____

21. $-2.50 -4.00 \times 180$ power at the 45 _____ and 135 _____

22. $-3.00 +2.00 \times 180$ power at the 30 _____ and 90 _____



Write the Rx in the _____ x20
 _____ x110

What would be the power in the X 20 _____
 X 155 _____
 X 80 _____
 X 170 _____
 X 110 _____

What base curve would you recommend for this Rx?
 $-2.50 -0.25 \times 180$

WORDS TO KNOW

Accommodation: Focusing of the eye on a near object through relaxation of the ciliary muscle and thickening of the lens.

Amblyopia: The loss of vision without any apparent disease of the eye.

Ametropia: A refractive error in which the eye, when in a state of rest, does not focus the image of an object upon the retina; includes hyperopia, myopia, and astigmatism.

Aniseikonia: A condition in which the ocular image of an object as seen by one eye differs so much in size or shape from that seen by the other eye that the two images cannot be fused into a single impression.

Anisometropia: A condition in which the refractive error of one eye significantly differs from that of the other; each eye must have the same sign, i.e. both eyes are myopic or hyperopic.

Antimetropia: The refractive condition whereby the signs in each eye are opposite; one eye is hyperopic while the other is myopic.

Aphakia: An absence of the crystalline lens of the eye.

Aqueous humor: The clear watery fluid that fills the anterior and posterior chambers within the front part of the eye.

Astigmatism: A refractive error that prevents the light rays from coming to a single focus on the retina because of different degrees of refraction in the various meridians of the eye.

Cataract: A condition in which the crystalline lens of the eye, or its capsule, or both, become opaque, with consequent loss of visual acuity.

Choroid: The vascular intermediate coat that furnishes nourishment to other parts of the eyeball.

Ciliary body: That portion of the vascular coat between the iris and the choroid. It consists of ciliary processes and the ciliary muscle.

Compound hyperopic astigmatism: The refractive error which results in two points of focus falling behind the retina.

Compound myopic astigmatism: The refractive error which results in two points of focus falling in front of the retina.

Cornea: The clear transparent portion of the outer coat of the eyeball forming the covering of the aqueous chamber.

Crystalline Lens: A transparent colorless body suspended in the front part of the eyeball, between the aqueous and the vitreous, the function of which is to bring the rays of light to a focus on the retina.

Diplopia: The seeing of one object as two.

Emmetropia: The refractive condition of the normal eye—when the eye is at rest, the image of distant objects is brought to a focus on the retina.

Fovea: A small depression in the retina at the back of the eye; the part of the macula adapted for most acute vision.

Fusion: The power of coordination by which the images received by the two eyes become a single image.

Glaucoma: An ocular disease having as its primary characteristic a sustained increase in intraocular pressure that the eye cannot withstand without damage to its structure or impairment of its function.

Hyperopia: A refractive error in which, because the eyeball is short or the refractive power of the lens is weak, the point of focus for rays of light from distant objects falls behind the retina; accommodation to increase the refractive power of the lens is necessary for distance vision as well as near vision.

Iris: The colored circular membrane suspended behind the cornea and immediately in front of the lens. The iris regulates the amount of light entering the eye by changing the size of the pupil.

Irregular Astigmatism: A refractive condition caused by a cornea which is damaged and irregular so that rays of light come to many focal points on the retina; is not correctable by cylinders.

Iseikonic Lenses: Spectacle lenses specially designed to manipulate image size when dissimilar sizes makes it difficult or impossible for fusion to occur; such as in aniseikonia.

Macula: The small area of the retina that surrounds the fovea and that with the fovea, comprises the area of the retina that gives distinct vision.

Mixed Astigmatism: The refractive condition in which light comes to two points of focus where one point is in front of the retina, and the other is behind the retina.

Myopia (nearsightedness): A refractive error in which the eyeball is too long in relation to its focusing power; thus the point of focus for rays of light from distant objects is in front of the retina.

Oculus Dexter (O.D.): The right eye.

Oculus Sinister (O.S.): The left eye.

Oculus Uterque (O.U.): Each eye.

Optic Nerve: The special nerve of the sense of sight that carries impulses from the retina to the brain.

Phoria: A root word denoting a latent deviation in which the eyes have a constant tendency to turn from the normal position for binocular vision; used with a prefix to indicate the direction of such deviation (for example, hyperphoria, esophoria, exophoria).

Presbyopia: A gradual lessening of the power of accommodation due to a physiologic change that becomes noticeable about the age of 40 years.

Ptosis Crutch: A small spring wire attached nasally to a spectacle frame to lift a drooping lid.

Pupil: The central opening of the iris through which light is permitted to enter the eye.

Regular Astigmatism: A refractive condition in which rays of light come to two focal points and is correctable by cylinders.

Retina: Innermost coat of the eye, formed of sensitive nerve elements and connected with the optic nerve.

Rods and Cones: Two different kinds of cells that form a layer of the retina and act as light-receiving media.
Cones are concerned with visual acuity and color discrimination; whereas rods are employed for motion and vision at low degrees of illumination (night vision).

Sclera: The white part of the eye—a tough covering which, with the cornea, forms the external protective coat of the eye.

Scotoma: A blind area of reduced vision in the visual field.

Simple hyperopic astigmatism: The refractive condition where one point of focus falls on the retina and the other point of focus falls behind the retina.

Simple myopic astigmatism: The refractive condition where one point of focus falls on the retina and the other point of focus falls in front of the retina.

Strabismus: Failure of the two eyes to simultaneously direct their gaze at the same object because of muscle imbalance. Also known as squint.

Suspensory Ligaments: The suspensory apparatus of the lens. Numerous fine tissue strands that stretch from the ciliary processes to the lens equator and support the lens in place.

Tropia: A root word denoting an obvious deviation from normal of the axis of the eyes; used with a prefix to denote the type of strabismus (for example, heterotropia, esotropia, exotropia).

Vitreous Humor: The transparent, colorless mass of soft, gelatinous material filling the eyeball behind the lens.

"A" Measurement In the Box Measurement System is the longest horizontal dimension of the eye wire of a frame. The measurement includes the groove in the eye wire of the bevel of a lens sized and shaped for the frame.

Abbe Value It is the reciprocal of the dispersive power of a lens material. It is a scalar used to compare the chromatic properties of lenses of different indices. Also referred to as the *Nu value*.

Accommodation Increase in optical power by the eye in order to maintain a clear image (focus) as objects are moved closer. Occurs through a process of ciliary muscle contraction and zonular relaxation that causes the elastic-like lens to "round up" and increase its optical power. Natural loss of accommodation with increasing age is called presbyopia.

Add The power that a lens segment has in addition to that power already present in the distance portion of the lens.

Amblyopia Also known as "lazy eye." Although there are different types of amblyopia, the term generally references a condition in children, in which the vision in one eye does not develop properly and the child cannot see well with it.

ANSI Standards ANSI is an acronym for American National Standards *Institute*. This organization has established guidelines for many different types of products. In the ophthalmic business there are three standards we are interested in. They are:

Z80.1 - This provides guidelines assessing the quality of single vision and multifocal spectacle lenses. It also includes information about methods for testing the impact resistance of prescription lenses.

Z80.3 - This provides guidelines for assessing the quality of plano sunglass lenses. This standard includes information about methods for testing the impact resistance of sunglass lenses.

Z87.1 - This provides guidelines for assessing the quality and make-up of industrial safety spectacles. This standard includes references about methods of testing impact resistance of lenses and safety frames.

Anisometropia A condition of the eyes in which they have unequal refractive power.

Anterior Chamber The space between the iris and the cornea.

Anti-Reflective Coating A chemical coating applied to eyeglass lenses to reduce surface reflections.

Aperture The opening in an ophthalmic frame into which a lens is inserted. Aperture dimensions (in millimeters) do not include the depth of the bevel groove. That principal meridian which contains only the spherical power component of a spherocylinder lens.

Aphakia The absence of the lens of the eye.

Aqueous The clear, watery fluid contained in the space between the cornea and iris of the eye. Flows between and nourishes the lens and cornea. Secreted by the ciliary processes.

Aspheric surface A nonspherical surface curvature commonly used to improve optical performance, particularly for high refractive powers. Such curvatures are often derived from the oblique intersection of a plane and a conical surface, and are referred to as "conoids" or "conic" sections.

Asthenopia Eyestrain.

Astigmatism The presence of two different curves on a single refracting surface or within the eye causing light to focus as two line images instead of a single point.

Axis The axis of a lens is the line which goes through the optical center. When applied to a cylinder or toric lens is the meridian that contains the spherical power and is 90 degrees from the cylinder power. If, for example, the axis of a minus cylinder prescription is at 180 degrees, the most minus power is in the 90th meridian.

"B" Measurement In the Box Measurement System it is the longest vertical dimension of the eye wire of a frame. The measurement includes the groove in the eye wire of the bevel of a lens sized and shaped for the frame.

Barrel The housing for a screw on a pair of eyeglasses.

Base Curve With rare exceptions the Base Curve of a lens is the flattest or lowest powered curve (front surface of a single vision or multifocal lens. On a toric surface, the base curve is the flattest or lowest powered curve.

Best Form Lens A lens designed with a base curve chosen to control oblique astigmatism and mean oblique error. The more commonly used term is corrected curve lens.

Bevel apex The point on the bevel of a lens.

Bifocal Lenses of eyeglasses that have two distinct areas of focus. The prescription in the top portion helps to see distant objects while the prescription in the bottom portion of the lens helps to see near objects.

Binocular pupillary distance The measurement between the patient's pupils, expressed in millimeters. Also called IPD or PD.

Binocular vision The blending of the separate images seen by each eye into a single image going to the brain, allowing three dimensional images, which leads to depth perception.

Block A device that is attached to the surface of a lens to hold it in place during the edging or surfacing process.

Blocker A device used to place a block on a lens in order to hold it in place during the edging or surfacing process.

Box-O-Graph A device containing grids and slides used in the measurement of patterns and edged lens size.

Box Measurement System A system for specifying the eye size and bridge size of a frame based on the dimension box drawn around the lens or eye wire where each of the four sides of the box are tangent to the bevel of the lens or groove in the eye wire. It includes the "A" measurement, "B" measurement, Distance Between Lenses, and the Effective Diameter of the eye wire.

Boxing system A system of lens measurement based upon the enclosure of a lens by horizontal and vertical tangents that form a box or rectangle.

Bridge The supportive structural member connecting the two eyes of an ophthalmic frame front.

Bridge size The shortest horizontal distance between lenses (DBL), measured in millimeters.

Carrier The optically unusable outer portion of a lenticular lens that "carries" the optically usable central portion.

Center thickness The thickness of a lens at the major reference point.

Centration The act of positioning a lens for edging such that it will optically conform to prescription specifications.

Choroid The layer of the eye that is filled with blood vessels and nourishes the retina.

Chromatic Aberration The characteristic of a lens material that causes white light to break up into its component colors. Shorter wavelengths focus closer to the lens than the focal point and longer wavelengths focus farther away.

Ciliary Body Tissue inside the eye made up of ciliary muscles and ciliary processes.

Ciliary Muscles The muscles that relax the zonules to enable the lens to change shape for focusing at distance or near.

Ciliary Processes The extensions or projections of the ciliary body that secrete aqueous humor.

Coma An aberration that results in a point object being transformed into a comet-like patch in the image. It is the manifestation of unequal magnification in different zones of a lens for obliquely incident rays of light emanating from an off axis object.

Comascope An instrument that utilizes polarized light to show strain patterns in glass or plastic.

Combination frame A frame whose front consists of a metal chassis with attached trim parts (sometimes known as top rims). These trim parts are typically plastic, aluminum or other metal, and are attached to the top portion of the chassis. These top rims may serve functional or cosmetic purposes, or both.

Concave An inward-curved surface.

Conjunctivitis (Pink Eye) Inflammation of the conjunctiva.

Convergence (1) An inward turning of the eyes as when looking at a near object.
(2) The action of light rays traveling toward a real image point

Convex An outward curved surface.

Coolant A recirculating liquid used to cool and lubricate the lens/grinding wheel interface during the grinding process.

Corrected curve lens A lens that has been designed to reduce peripheral power errors for the conditions of intended use over a specified portion of the field of view.

CR-39 A registered trademark of Pittsburgh Plate Glass Company for an optical plastic known as Columbia Resin 39.

Cribbing The process of reducing a lens blank to a smaller size in order to speed the surfacing process.

Cross curve The maximum surface power of a toric surface (90 degrees from the base curve meridian).

CRP Abbreviation for corneal reflection pupillometer, the instrument used to measure corneal reflection.

Cylinder A lens having refractive power in one meridian only and used in correction of astigmatism.

Cylinder axis An imaginary reference line used to specify cylinder or spherocylinder lens orientation corresponding to the meridian perpendicular to that of maximum cylinder power.

Cylinder lens A special case of the spherocylinder lens in which one of the principal meridians has zero refractive power.

Cylinder power The difference (plus or minus) between the powers measured in the two principal meridians of a lens.

DBC Abbreviation for Distance Between Centers.

DBL Abbreviation for Distance Between Lenses.

Decentration The lateral distance between the mechanical center and the optical center of the lens. It is a function of the frame PD and the patient's PD.

Density Measurement of weight based on a certain amount of material. The weight of a lens material is reflected as its density.

Diopter A unit of lens refractive power equal to the reciprocal of the lens focal length in meters.

Diplopia The seeing of one object as two.

Distance Between Centers (DBC) Distance between the geographic center of the eye wire. Commonly referred to as the Frame PD.

Distance Between Lenses (DBL) The measurement between the nasal eye wires where they are closest together. Frames that are Box Measured frames use this number to specify the bridge size.

Distortion Distortion is caused by magnification of an extended object for rays parallel to the axis that varies with the distance of the ray bundle from the axis. This aberration affects the shape rather than the sharpness of the image. Lenses that minimize distortion are called orthoscopic.

Dress To resharpen the cutting surface of a grinding wheel.

Dress frame A frame for prescription or corrective lenses, intended for ordinary use in correcting or improving vision. Such a frame is not intended for occupational or safety use.

Drop Ball Test This is a test prescribed by the FDA in regulation CFR21 801.410. The test requires that the lens be subjected to the impact of a 5/8" steel ball dropped from a height of 50".

ED Abbreviation for effective diameter.

Edged lens A lens whose periphery has been ground (flat, beveled or grooved) to a specific size and shape.

Edger That piece of machinery used to physically grind the uncut lens blank to the shape and size of the frame.

Effective diameter (1) Defines the minimum lens that will fit a frame when the geometric center of the lens is exactly centered in the frame. (2) A linear measurement, expressed in millimeters, equal to twice the longest radius of an ophthalmic lens measured from its geometric center to the apex of its edge.

Effective Power In general terms it is the ability of a lens to focus parallel rays of light. In a lens it is the back vertex power.

Esophoria (Convergent Squint) A tendency of one eye to turn inward.

Exophoria (Divergent Squint) A tendency of one eye to turn outward.

Esotropia A manifest inward deviation of one eye.

Exotropia A manifest outward deviation of one eye.

Eye Size Number printed on the frame that represents a size code for that eye wire. Frames that are Box Measured Frames give the "A" measurement as the eye size.

Eyewire The rim of the frame that goes around the lenses.

Face Form The curvature of the front of an ophthalmic frame.

Finished blank A lens having both front and back surfaces ground to the desired powers but not edged to the shape of the frame.

Fitting Triangle The triangle formed by connecting the three points of contact of a frame on the head. The three points are the crest of the nose and the top of each ear.

Flat Lens A lens with one surface whose radius is infinite (plano).

Focimeter An instrument used to determine vertex power, axis location, optical center, and major reference point location and prism power at a given point on an ophthalmic lens.

Former The British equivalent of a pattern.

Front A component of an ophthalmic frame, typically consisting of a bridge and eyewires.

Fused lens A multifocal lens made from three or more pieces of glass that are thermally sealed together. On fused multifocal lenses, curve changes or ledges are invisible.

GC Abbreviation for geometric center.

Generating The process of rapidly cutting the desired surface curvature onto a semi-finished lens blank.

Geometric center The intersection of the horizontal and vertical centerlines of a box that circumscribes the lens shape.

Glazing The insertion of lenses into a spectacle frame. (2) The clogging of empty spaces between the exposed abrasive particles of an abrasive wheel resulting in reduced grinding ability.

Glazed Assembled with appropriate ophthalmic lenses.

Hand edger A grinding wheel especially for grinding lenses by hand.

Hinges Part of the hardware of the frame. The hinges attach to both the temple and front of the frame.

High index lenses Lenses made with a higher refractive material than CR-39 plastic or glass. High Index Lenses are thinner than lower index lenses.

Impact Resistance In the regulatory sense it is the ability of an ophthalmic lens to remain intact while being subjected to the drop ball test. In a more global sense, it is simply the ability of a lens to withstand the strain (remain intact) resulting from an applied stress.

Inset The lateral separation between the distance optical center and the segment optical center. It is a function of the patient's distance and near PD.

Interpupillary distance The linear distance between the fixation axes of the wearer's two eyes. It is commonly referred to as the distance between the centers of the pupillary openings with the eyes focused on a distant object.

IPD Abbreviation for interpupillary distance.

Index of refraction A measure of the ability of a lens material to refract a ray of light of a given wavelength. This is usually stated for the wavelength of the helium d-line (587.56 nanometers). The higher the index, the more the refractive power of the lens. For the ophthalmic glasses most commonly in use, $n=1.5230$. The index of refraction for allyl resin, a plastic most commonly used for eyeglasses, is typically 1.4975.

Intermediate That area in a trifocal lens or blank which has been designed to correct vision at ranges intermediate to distant and near objects.

Keyhole bridge A bridge design for a front that does not permit continuous contact between the nose and the front in the area of the nasal crest.

Laminated lens A lens constructed as a sandwich of multiple layers of glass or plastic, or both, bonded together to form a single unit.

Lap A tool having a curvature matching that of the surface desired for a lens surface. The lens surface is rubbed across the face of the tool and, with the aid of pads, abrasives and polishes, the lens surface is brought to optical quality.

Layout The process of preparing a lens for blocking and edging.

LEAP A 3M Company adhesive pad blocking system.

Lens Aberration Defect in the image forming properties of a lens which results in an alteration in the quality, size, and shape of the image.

Lens pattern A cam, or template, used in lens edging equipment to generate the correct peripheral shape and geometric center location. Also called a lens former.

Lens protractor A millimeter grid on a 360-degree protractor used in the lens centration process for both surfacing and finishing.

Lens, single-vision A lens designed to provide correction for a single viewing distance.

Lens, spherical A lens that has the same refractive power in all meridians. Such a lens may have rotationally symmetrical aspheric (reflective) surfaces.

Lens, sphero-cylinder A lens that has different refractive power in the two principal meridians. It is often referred to as an astigmatic or toric lens and sometimes *incorrectly* referred to as a cylinder lens.

Lens, stock A lens supplied by a manufacturer with both surfaces finished and a specific back vertex power or powers. Such a lens has yet to be edged to a specific shape. Also known as factory finished, uncut.

Lens, toric A lens that has two distinct curvatures, at right angles (90 degrees) to each other.

Lens, uncut A lens with finished optical surfaces on both sides but not edged for mounting in a frame.

Lens measure An instrument that is used to measure surface curvature. Also called sagitta gauge, or lens clock.

Lens size The horizontal box dimension (A-dimension) of a finished lens. Also called eye size.

Lenticular A high-powered lens with the desired prescription power found only in the central portion. The outer carrier portion is ground so as to reduce edge thickness and weight in minus prescriptions and center thickness and weight in plus prescriptions.

Lenticular lens A lens, usually of a strong refractive power, in which the prescribed power is provided over only a limited central region of the lens, called the lenticular portion. The remainder of the lens is called the carrier and provides no refractive correction but gives dimension to the lens for mounting.

Lensometer The trade name for an instrument used for finding power and prism in spectacle lenses.

Library temple The style of temple that has almost no bend over the ear; it was originally designed for ease of removal.

Line, geometric center A horizontal line running through the geometric center of a lens. Also called normal mounting line.

MPD Abbreviation for monocular pupillary distance.

Major Reference Point (MRP) The point on the lens which provides all of the required prescription parameters including prism.

Marked surface power The nominal curve of a semi-finished lens marked in diopters, as expressed by the manufacturer. The difference between marked and actual tool curve of the surface represents the manufacturer's compensation for that base curve. This compensation allows standard tooling to be utilized over a range of prescriptions with little or no further compensation required by the laboratory to produce accurate vertex powers.

Marker A centering device used to accurately position a lens and stamp it with horizontal and vertical reference lines for use in accurate lens blocking.

Mechanical Center This is the point about which the lens rotates while being edged. It is synonymous with the Geometric Center of the eye wire.

Meniscus Lens A lens with a convex and a concave surface.

Meridian The line of intersection of a surface with a plane perpendicular to that surface at a specified point. When applied to a lens, it also may be defined as a plane that contains the optical axis.

Mineral lens A glass lens.

Minimum Blank Size (MBS) The smallest lens blank that can be used to properly shape and size a lens for a given frame. In mathematical terms, $MBS = \text{Effective Diameter} + 2(\text{Decentration per eye})$.

Minus cylinder form The form a prescription takes when written such that the value of the cylinder is expressed as a negative number.

Minus cylinder form lens A lens ground such that it obtains the cylinder power from a difference in surface curvature between two back surface meridians.

Minus lens A lens having negative dioptric power. It is thinner at the center than at the edge.

Molded blank A blank that is unfinished on both sides when it arrives from the factory. It is used to grind lenses for non-standard prescriptions or prescriptions that are particularly strong.

Monocular pupillary distance (Monocular PD) The measurement from the center of the nose to the pupil. Also called MPD.

Mounting (1) A rimless or semi-rimless spectacle frame. (2) The attaching of lenses to a rimless or semi-rimless spectacle frame.

Multifocal lens A lens designed for two or more viewing ranges, for example, bifocal or trifocal lenses.

Myodisc Synonym for negative lenticular.

Nasal That side of the lens of frame that is towards the nose.

Neutralize To determine the refractive power of a lens.

Oblique Astigmatism Aberration that occurs when a ray bundle from an off-axis object point passes obliquely through a spherical surface. The rays of light do not form a point but, rather, an Interval of Sturm with two foci (tangential and sagittal) and a circle of least confusion. The foci and interval are formed as though the surface were toric. Also referred to as Marginal Astigmatism.

OD Latin, oculus dexter (right eye).

One-piece multifocal lens A multifocal lens or a blank fabricated from a single piece of glass or plastic.

Optical Center (OC) The point on a lens where rays of light are not displaced. It is the thinnest point on a minus lens and the thickest point on a plus lens.

Optically centered pattern A pattern with its mechanical center above boxing center.

Optyl The trade name for an epoxy resin material used to make spectacle frames.

OS Latin, oculus sinister (left eye).

Overplussed Too much positive power.

Pantoscopic Angle Technically this is the angle formed by a plane perpendicular to the temple of a frame and the plane of the eye wires.

Parallax The apparent change in position of an object as the result of a change in the viewing angle.

Pattern A plastic or metal piece having the same shape as the lens aperture for a given frame. Used in lens edging as a guide for shaping the lens to the frame.

Periscope Lens A lens that has a 1.25D base curve. The base curve is on the front surface of minus prescriptions and on the back on plus.

Photochromic lens A lens that darkens in response to the ultraviolet component of sunlight.

Plano lens A lens that has zero refractive power.

Plano surface A flat surface having zero surface power, or an infinite radius of curvature.

Plus cylinder form The form a prescription takes when written such that the value of the cylinder is expressed as a positive number.

Plus cylinder form lens A lens ground such that it obtains its cylinder power from a difference in surface curvature between two front surface meridians.

Plus lens A lens that has positive refractive power. It is thicker at the center than at the edge.

Polycarbonate lenses Polycarbonate Lenses are the most durable of all lenses. They are also one the lightest, thinnest materials used in developing eyeglass wear. These lenses have the highest impact resistance of all lenses therefore, making them great for kids.

Power cross A schematic representation upon which the two major meridians of a lens or lens surface are depicted.

Precoat A spray or brush-on liquid that when applied to a lens, protects the surface during processing, and/or makes the adhesion of a block to the lens possible.

Prentice's Rule According to this formula, the amount of prism generated by a lens is a function of the power of the lens and the distance of the rays of light from the optical center of the lens.

Principal meridians The two mutually perpendicular meridians of a spherocylinder lens or toric optical surface with minimum and maximum power.

Prism A prism lens is designed to displace the image without effecting focusing the rays of light.

Progressive Lens A multifocal lens on which the additional reading power is generated via a gradual increase in power which is facilitated by a transition zone.

Prentice's Rule A rule that states that the decentration of a lens in centimeters times the power of the lens is equal to the prismatic effect.

Prism diopter A unit of measurement used to express the angle of deviation of a ray of light by a prism or lens. In these units, prism power is measured in centimeters as the displacement of a light ray perpendicular to its line of incidence at a distance of one meter.

Propellor A misalignment of the eye wires of a frame due to a twisting of the front about the bridge bar. This results in a condition in which the lower eye wire on one side is closer to the cheek and farther away on the other side.

Reading glasses Reading Glasses have lenses with stronger powers because they incorporate both the distance and near powers to concentrate power for easy reading for the wearer.

Refraction The bending of light rays caused by prisms and lenses.

Refractive index The ratio of the speed of light in a medium (such as air) to the speed of light in another medium (such as glass).

Refractive power The ability of a lens or an optical surface to produce a change in the convergence or divergence of a beam of light, usually expressed in diopters.

Riding bow temple A style of temple that bends around to hug the ear. It is particularly useful to people whose jobs are very active, or for small children.

Rimless frame A type of frame that provides no, or only partial, peripheral support for the lenses.

Saddle bridge A bridge design for a front that permits continuous contact between the nose and the front in the area of the nasal crest.

Safety bevel To remove the sharp interface between the lens surface and bevel surface and the sharp point at the bevel apex.

Seg depth The longest vertical dimension of a lens segment.

Seg drop The vertical distance from the major reference point to the top of the seg when the seg is lower than the MRP.

Segment A specified area of a multifocal lens having different power from the major portion. This also may refer to the actual piece of material added to the lens in the case of a fused or cemented multifocal lens. Also called the addition.

Segment Below The vertical separation between the distance optical center and the top of the segment. It is function of the vertical dimension of the eye wire and the seg height.

Segment Height Top of segment to the lowest point on the lower eye wire.

Semi-finished blank A lens with only one side having the desired curvature. The second side must be surfaced in order to bring the lens to the desired power and thickness.

Set number The compensating number used with a pattern to arrive at a compensated eyesize setting for the edger.

Sizer A frame chassis or frame front used exclusively for checking edged lens size accuracy.

Skull temple A style of temple that has a slight bend that allows the frame to fit easily over the ear, and to hug the head lightly.

Spatula temple Another term for skull temple; also called paddle temple.

Spectacle An ophthalmic device consisting of two ophthalmic lenses and a supporting frame to position and retain the lenses in proper optical alignment with the eyes.

Sphere A lens having a single refractive power in all meridians.

Sphere power In a spherical lens, the dioptric power of the lens. In a spherocylinder lens, the sphere power is located in the cylinder axis meridian.

Spheric surface A curved surface having the same radius of curvature in all meridians.

Spherical Aberration The inability of a lens to focus axial and paraxial ray bundles that are parallel to the axis at point in image space. The result is that peripheral rays focus closer to the lens than more central ray bundles.

Spherocylinder The combination of a sphere lens and a cylinder combined into a single lens.

Spotting The placing of spots on a lens correctly oriented for axis in order to indicate the location of the major reference point and horizontal meridian.

Stars Microchips at the lens surface/lens bevel interface.

Surface power (R) The refractive power of a lens surface having index of refraction (n) is a measure of its ability to refract light, and is expressed in diopters. The expression relating (R) to (S) and (n) is: $R = [(n-1)S]/0.530$. Since common ophthalmic materials do not have indices of refraction equal to 1.530, there is not a one-to-one correspondence between surface tool power (S) and surface refractive power (R). For example, common ophthalmic

glass has an index of refraction of 1.5230. Therefore, a one-diopter surface tool will produce a surface refractive power (R) of 0.987 diopter. Surface refractive power is also called true power.

Surface tool power (S) The actual radius of curvature of a tool or the surface it produces. By common usage in the United States, a tool with a radius of curvature of 530mm will produce a surface tool power (S) of one diopter. When (r) is the radius of curvature in millimeters, then $S = 0.530/(0.001r)$.

Surfacing The process of creating the prescribed refractive power, prism, and major reference point location on a lens by generating the required curves and bringing the surface to a polished state.

Swarf The accumulated waste material present as a result of the grinding process.

Temple Angle The angle between the front and the temple of a frame. Temple angle can be observed when looking at the frame from above.

Temple screws Tiny screws that connect the front and temple halves of the hinges.

Temple Size Total length of the temple from the hinge to the end in millimeters or inches.

Toric surface A surface in the form of a torus having different powers in two principal meridians. The shape may be visualized as that of a section cut from a doughnut or from a football-shaped surface.

Total Inset Lateral separation between the Mechanical Center of the lens and the near optical center.

Transposition Changing the powers of a sphero-cylinder lens or astigmatic prescription from one cylinder form to the other (- to - or - to -).

Trifocal lenses Lenses having three areas of viewing, each with own focal power. Usually the upper portion is for distance viewing, the lower for near, and the middle portion for distances in between.

Uncut Lens A lens which has prescription that has not been edged for the frame.

V Bevel A lens edge configured to have the form of a V across the whole breadth of the lens edge.

Vertex depth The depth of the surface curve on a lens measured over a specific diameter. Also called the sagitta.

Vertex power The inverse of the distance, expressed in meters, from the lens vertex to the corresponding focal point. This is expressed in diopters. In a prescription, the spherical component of power and cylindrical component are always expressed in terms of rear (or back) vertex power. Focimeters are designed to measure vertex power directly.

Wave A local ripple-like irregularity in a lens surface.

Zyl Cellulose acetate, used to make frames.

Zyl frame A frame made from cellulose acetate.